

RIYADH

HeatReady

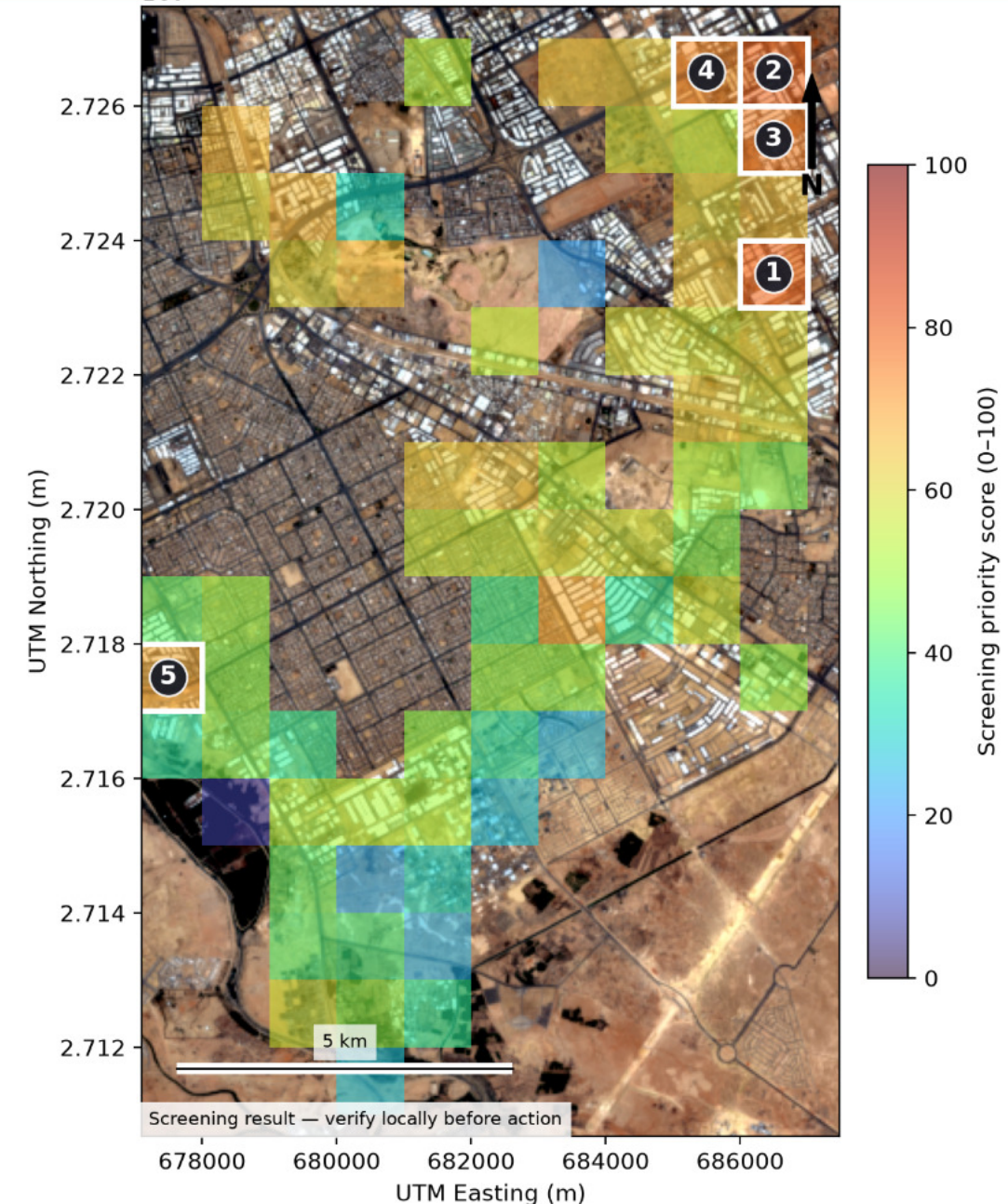
A growth-to-heat priority map for southeastern Riyadh

One clear decision

Which 1 km zones should a municipal planning and heat-resilience team inspect first?

Milkaholics • Urban Expansion, Land Use Change & Heat Risk • United Arab Emirates

SATELLITE-DERIVED PRIORITY MAP



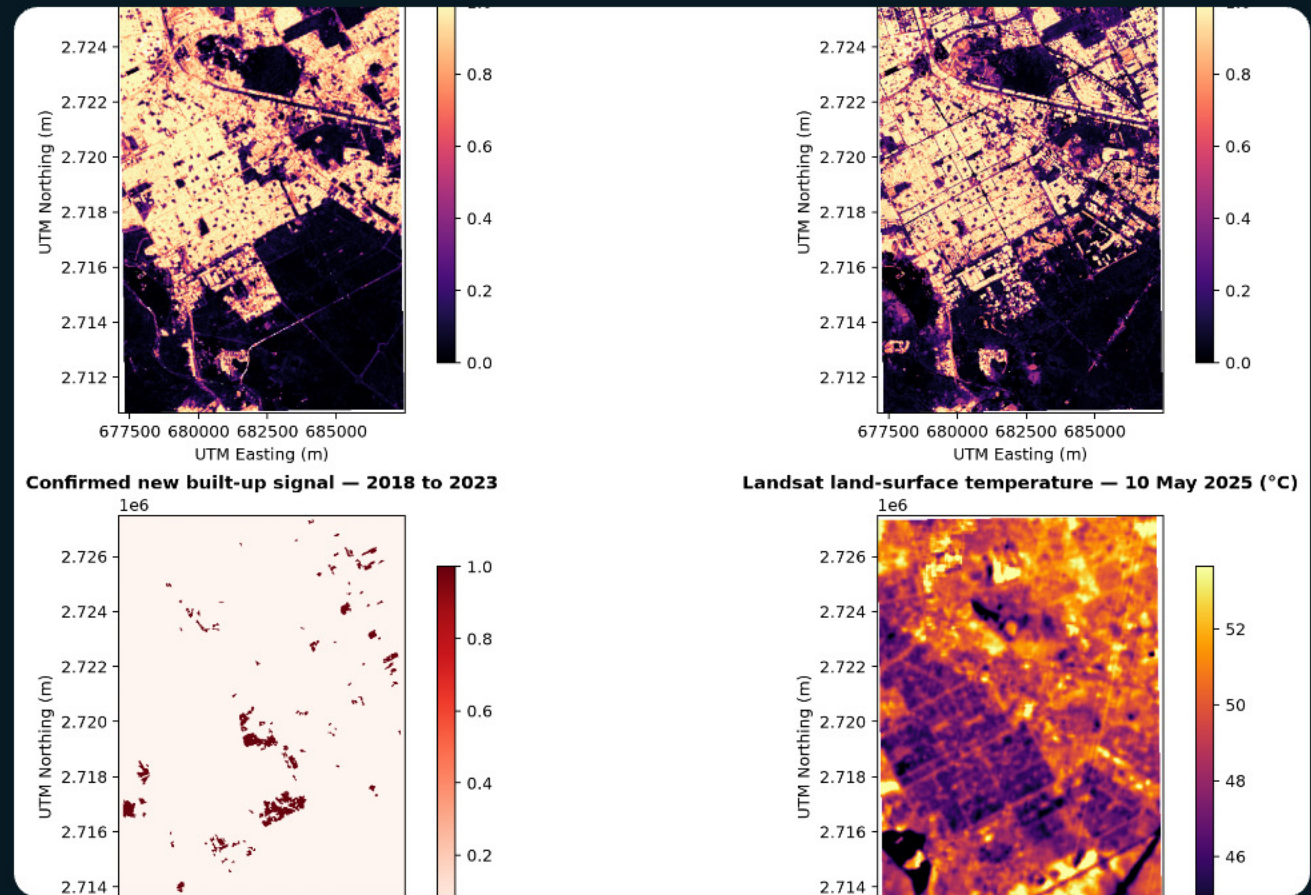
Growth and heat are visible - but the decision is fragmented

168.30 km² study area

Planners may need to compare separate growth, heat, population, and vegetation layers before choosing where to inspect.

Affected group

Residents and workers in fast-changing, hot urban-fringe zones.



One user. One decision. One testable PoC.

USER

Municipal planning + heat-resilience team

Reviews zones, local plans,
infrastructure, and field
conditions.



DECISION

Where should we inspect first?

Output: ranked 1 km cells with
the evidence behind each
score.



ACTION

Local review before intervention

Shade, trees, cool roofs, or
pavements are options to assess
- not automatic recommendations.

HONEST GAP

No municipal user interview has happened yet. The current workflow and value are hypotheses to test during incubation.

Every source has one job - and the result depends on EO

Sentinel-2

Optical reflectance, quality mask, vegetation + built features

Sentinel-1

Radar structure to reduce bright-desert confusion

Annual LULC

Six mapped years propose 2018-2023 growth

Landsat 8

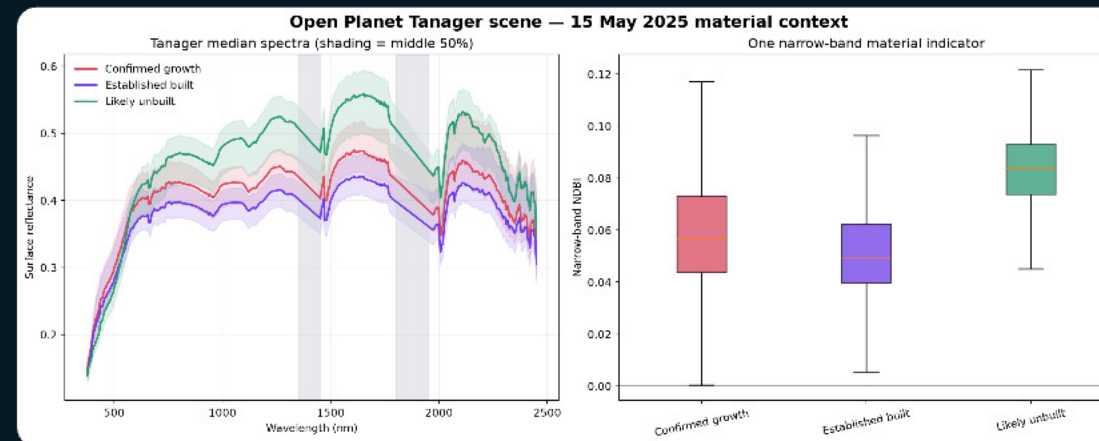
2025 land-surface temperature

WorldPop

Modelled 2025 population exposure

Planet Tanager

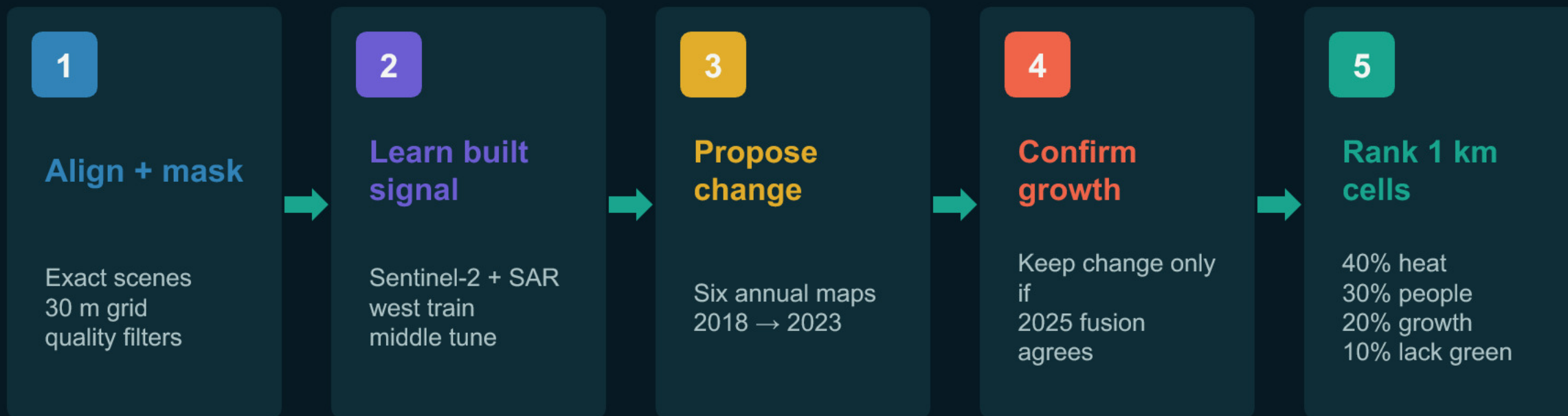
426-band 2025 spectra: material context + failure-mode test



Hyperspectral role

Tanager improves the analysis by testing a known desert failure mode. Likely-unbuilt surfaces had higher median narrow-band NDBI than established-built surfaces.

A conservative chain from open scenes to inspection priority

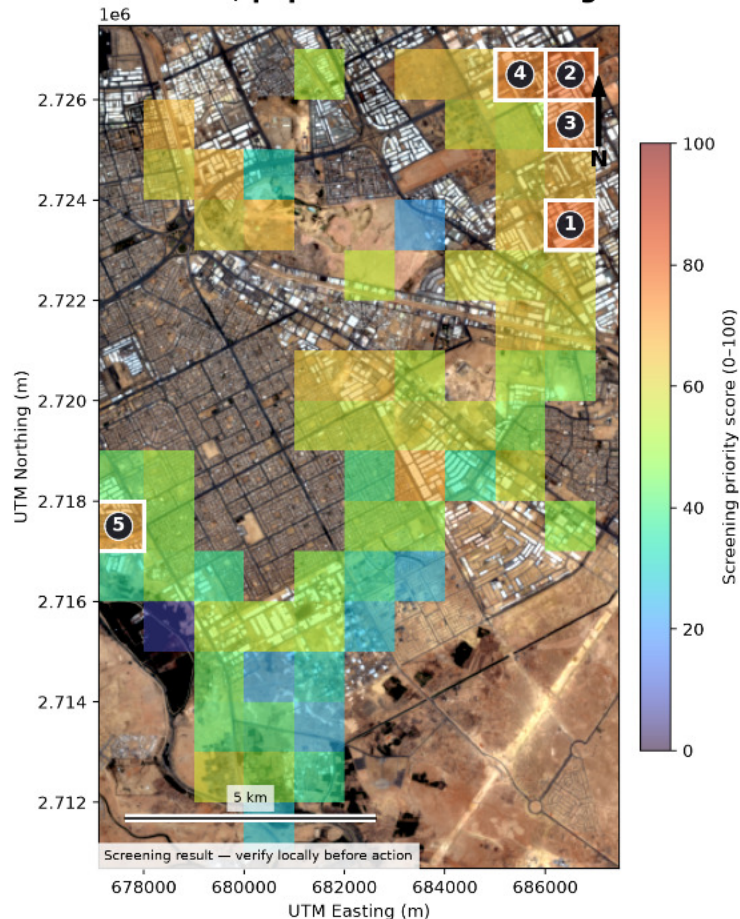


Validation design

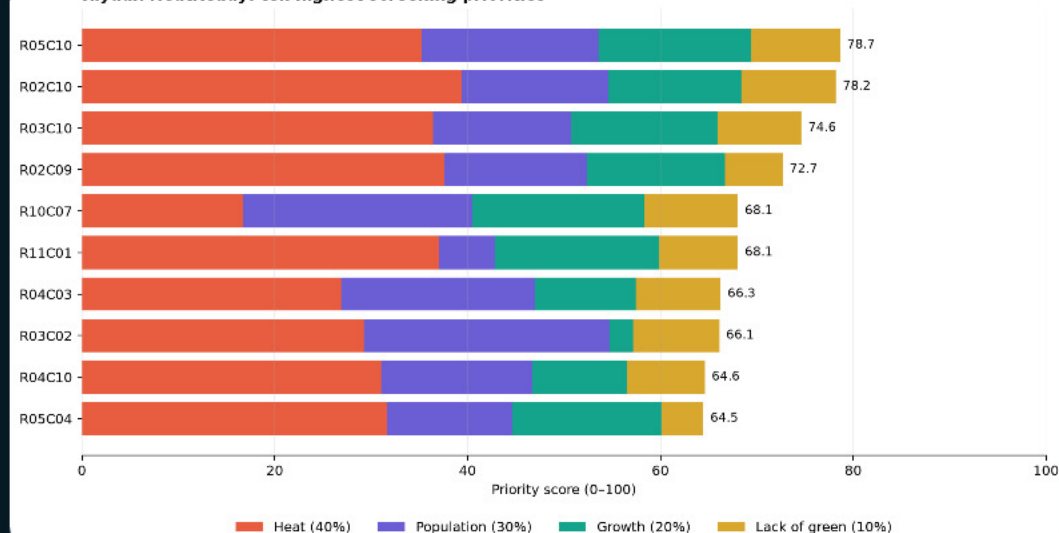
The east side is held out for testing against WorldCover. The final growth layer must pass two independent tests: annual change + optical-radar confirmation.

Six data sources become one ranked inspection list

Riyadh HeatReady decision map
Top zones where heat, population and recent growth overlap



Riyadh HeatReady: ten highest screening priorities



3.74 km²

confirmed growth
signal

14 / 67

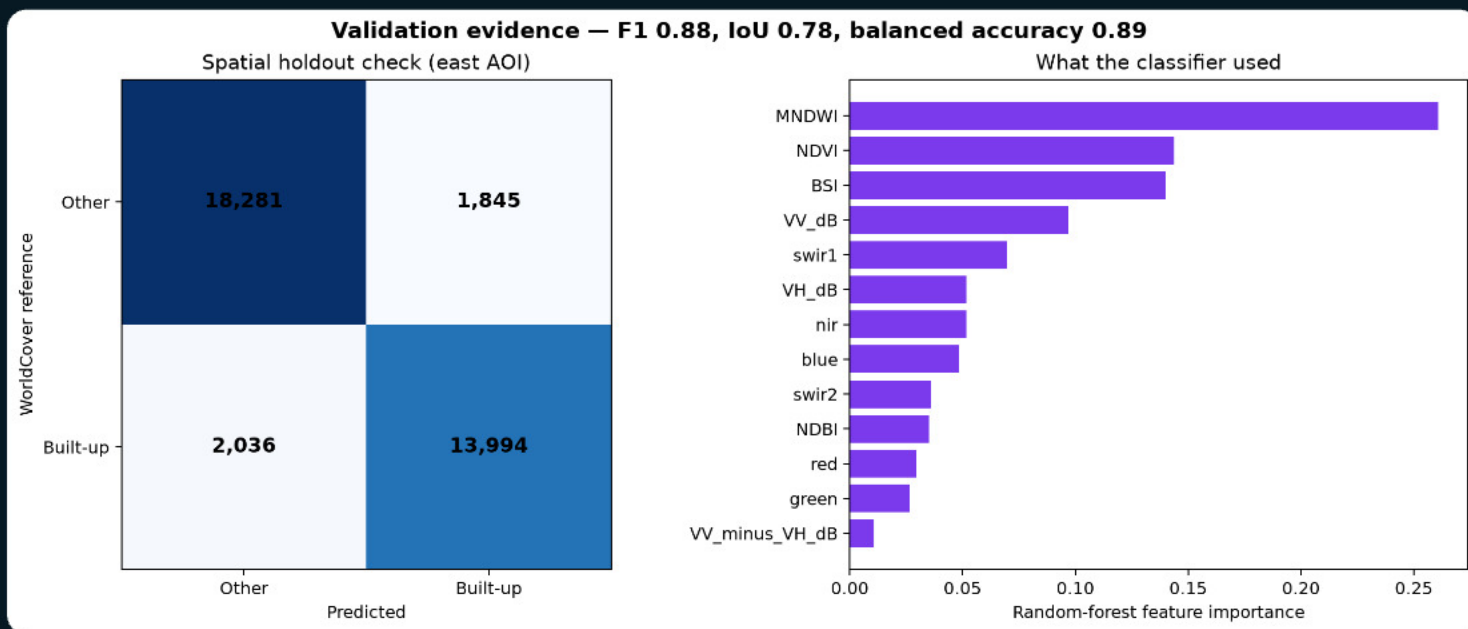
High / eligible cells

78.7

top score: R05C10

Screening result - verify locally before action

Strong satellite agreement - with clear boundaries on the claim



0.88

F1 agreement

0.78

intersection over union

Four limits to remember

- 1 WorldCover agreement is not field accuracy.
- 2 Surface temperature is not air temperature.
- 3 WorldPop is modelled, not a census.
- 4 Bright desert can still confuse built surfaces.

The 2022 mapped-area dip is useful evidence of uncertainty - not something to hide.

A repeatable screen can focus scarce local inspection time

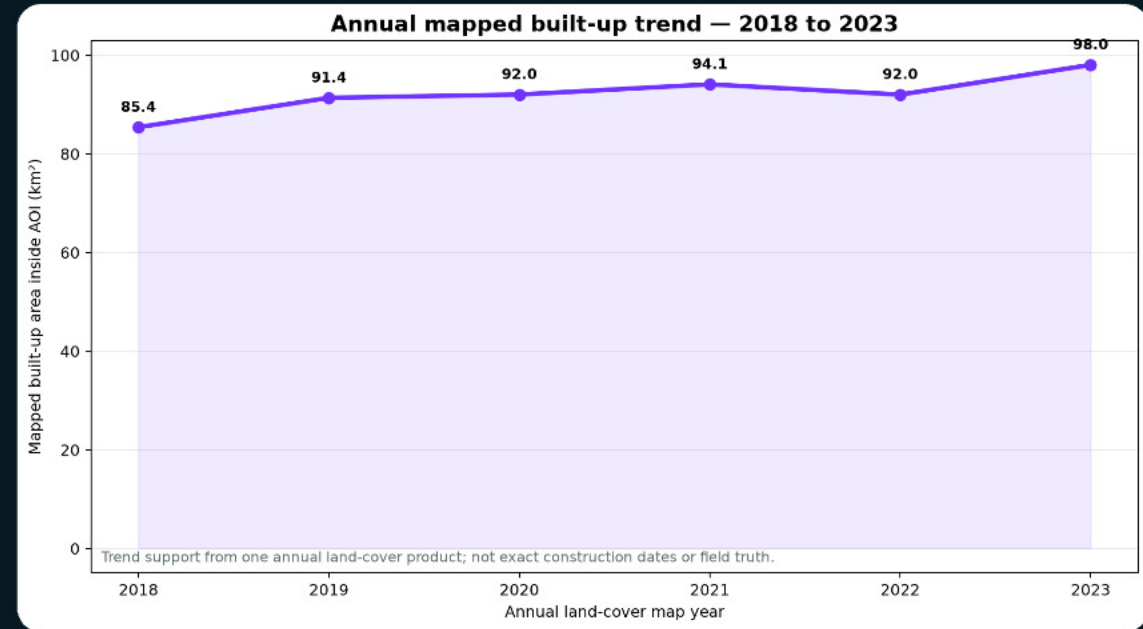
PILOT HYPOTHESIS

Inspect the top five zones

A planning team compares the ranking with local plans and field conditions, then records useful hits, false alarms, and time saved.

Deliverable

Refreshable dashboard + CSV
+ GeoJSON + GeoTIFF



85.39

km² mapped in 2018

98.03

km² mapped in 2023

1 area

tested; not all Riyadh

Turn a credible screen into a user-validated municipal tool

- 01 Interview** 3+ planners, GIS staff, or heat stakeholders
- 02 Agree** User decision, weights, thresholds, and success measure
- 03 Validate** Field-check top zones and label false alarms
- 04 Enrich** City plans, worker exposure, and higher-resolution data
- 05 Repeat** Run another Riyadh area and test update cost

Goal: prove usefulness with real users - then scale the data and product.